

UNIVERSITÀ
DEGLI STUDI
FIRENZE

B032426 (B255) - ARCHITECTURE, MODEL AND ANALYSIS OF CYBER PHYSICAL SYSTEMS

Bayesian Networks

Lecture 1

Prof. Francesco Flammini

francesco.flammini@unifi.it

UNIVERSITÀ DEGLI STUDI FIRENZE

General information

- Track on theory and application of Bayesian Networks for dependability modeling and analysis of cyber-physical systems
- Exam: evaluation and discussion of submitted project
- Lectures on Thursdays 10:30/11.00-13.00/13.30, Room 102 DIMAI
- Materials:
  - Lecture slides and links
  - Technical papers provided
- Reception time: on demand, email request
- Information and materials on course website (Moodle page)

UNIVERSITÀ DEGLI STUDI FIRENZE

Learning objectives of this lecture

At the end of this lecture you will be able to:

- Motivate the usage of Bayesian Network (BN) modeling formalisms in the context of dependability analysis of cyber-physical systems
- Define the BN modeling formalism
- Sketch simple BN models
- Explain some relevant BN properties
- Implement and evaluate simple BN models in automated tools

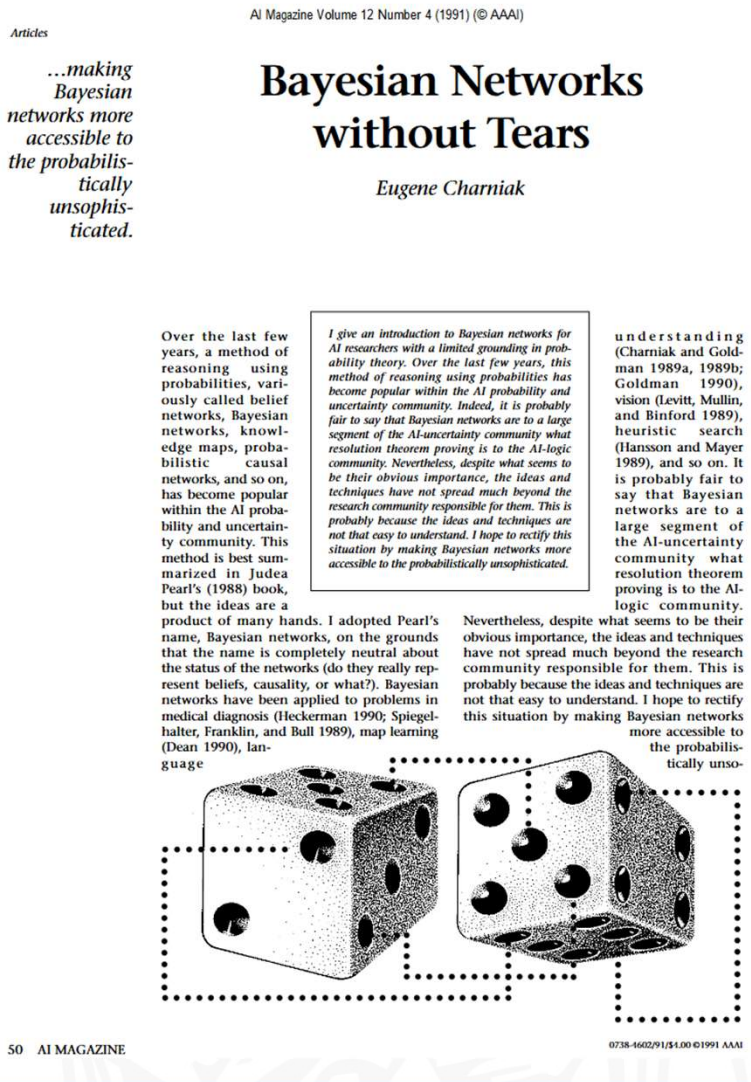
UNIVERSITÀ DEGLI STUDI FIRENZE

Learning materials for this lecture

Charniak, E. (1991), Bayesian Networks without Tears. AI Magazine, 12: 50-63.

https://doi.org/10.1609/aimag.v12i4.918

(open access)



AI Magazine Volume 12 Number 4 (1991) (© AAAI)

Articles

...making Bayesian networks more accessible to the probabilistically unsophisticated.

Bayesian Networks without Tears

Eugene Charniak

I give an introduction to Bayesian networks for AI researchers with a limited grounding in probability theory. Over the last few years, this method of reasoning using probabilities has become popular within the AI probability and uncertainty community. Indeed, it is probably fair to say that Bayesian networks are to a large segment of the AI-uncertainty community what resolution theorem proving is to the AI-logic community. Nevertheless, despite what seems to be their obvious importance, the ideas and techniques have not spread much beyond the research community responsible for them. This is probably because the ideas and techniques are not that easy to understand. I hope to rectify this situation by making Bayesian networks more accessible to the probabilistically unsophisticated.

Over the last few years, a method of reasoning using probabilities, variously called belief networks, Bayesian networks, knowledge maps, probabilistic causal networks, and so on, has become popular within the AI probability and uncertainty community. This method is best summarized in Judea Pearl's (1988) book, but the ideas are a product of many hands. I adopted Pearl's name, Bayesian networks, on the grounds that the name is completely neutral about the status of the networks (do they really represent beliefs, causality, or what?). Bayesian networks have been applied to problems in medical diagnosis (Heckerman 1990; Spiegelhalter, Franklin, and Bull 1989), map learning (Dean 1990), language understanding (Charniak and Goldman 1989a, 1989b; Goldman 1990), vision (Levitt, Mullin, and Binford 1989), heuristic search (Hansson and Mayer 1989), and so on. It is probably fair to say that Bayesian networks are to a large segment of the AI-uncertainty community what resolution theorem proving is to the AI-logic community. Nevertheless, despite what seems to be their obvious importance, the ideas and techniques have not spread much beyond the research community responsible for them. This is probably because the ideas and techniques are not that easy to understand. I hope to rectify this situation by making Bayesian networks more accessible to the probabilistically unso-

50 AI MAGAZINE

0738-4602/91/$4.00 ©1991 AAAI

UNIVERSITÀ DEGLI STUDI FIRENZE

Why Bayesian Networks?

- Relatively recent and powerful tool for reasoning under uncertainty
- Mostly known as probabilistic modeling tool within causal Artificial Intelligence (AI)
- The term **Bayesian network** was coined by Judea Pearl in 1985. In comparison: Fault tree analysis (FTA) was originally developed in 1962 at Bell Laboratories by H.A. Watson; Petri Nets (PN) first introduced by Carl Adam Petri in 1962 in his Ph.D. dissertation
- In dependability modeling and analysis, it has been shown that BN represent a good tradeoff between FT and PN in terms of expressive power, efficiency, and ease of use (see: FROM ARTIFICIAL INTELLIGENCE TO DEPENDABILITY: MODELING AND ANALYSIS WITH BAYESIAN NETWORKS, L. PORTINALE, A. BOBBIO, and S. MONTANI, Modern Statistical and Mathematical Methods in Reliability. October 2005, 365-381)

UNIVERSITÀ DEGLI STUDI FIRENZE

Further advantages of BNs

- They are especially suitable for diagnostic applications, which is very useful in failure identification and maintenance
- They can be used for multi-sensor information fusion leveraging on redundancy and diversity to improve the trustworthiness of event detection in monitoring/perception
- Compared to other powerful machine learning models that can be used for the same purposes, such as deep neural networks, BNs are intrinsically *transparent and explainable* and therefore much easier to assess in CPS certification processes
- Although simple in their basic formulation, they can be made more complex and extended as Time-Varying Bayesian Networks (TV-BNs) or Dynamic Bayesian Networks (DBNs)
- There exist algorithms for learning the parameters and structure of BNs from data

UNIVERSITÀ DEGLI STUDI FIRENZE

What is a Bayesian Network?

- First of all, it is a *graphical modeling formalism*, just like fault trees and Petri nets, which means is has *nodes* and *arcs* connecting them
- Being a «formalism», it has precise, mathematically defined, semiotics, syntax, and semantic (no representation ambiguities)
- Formally, it is defined as a *Directed Acyclic Graph* (DAG), whose nodes represent *stochastic variables*, and whose arcs represent *dependencies* among them, measured by *conditional probabilities*
- All those dependencies are quantified in the so-called *Conditional Probability Tables* (CPTs), which are embedded in the model

UNIVERSITÀ DEGLI STUDI FIRENZE

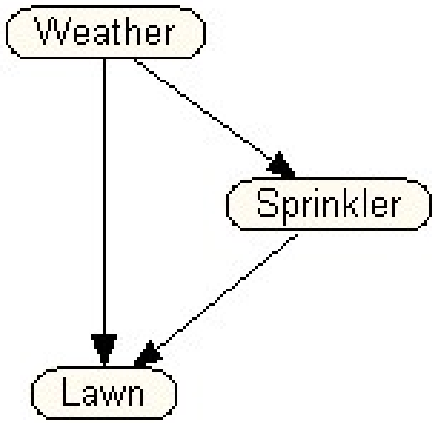
Simple BN example

- Weather has three states
  - Sunny, or
  - Cloudy, or
  - Rainy,
- Grass can be:
  - Wet, or
  - Dry,

Sprinkler can be:

On, or

Off.



Causal links:

- If it is rainy, this will make the grass wet directly.
- if it is sunny for a long time, that too can make the grass wet, indirectly, by causing us to turn on the sprinkler.

Example evaluations:

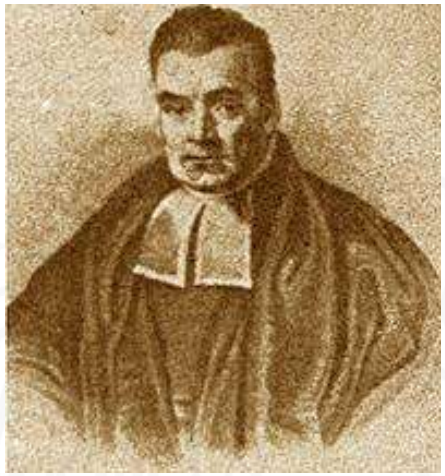
"if the lawn is wet, what are the chances it was caused by rain or by the sprinkler"

"if the chance of rain increases, how does that affect my having to budget time for watering the lawn".

UNIVERSITÀ DEGLI STUDI FIRENZE

Why «Bayes»?

Bayes nets are networks of relationships, hence nets. And they are named "Bayes" after Reverend Thomas Bayes, 1702-1761, a British theologian and mathematician who wrote down a basic law of probability which is now called Bayes rule.



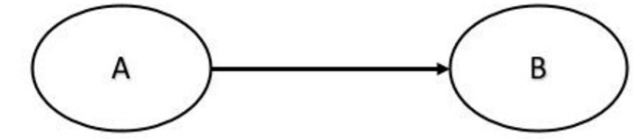
EXAMPLE
Suppose you live in London and you notice that during the winter, it rains 50% of the time and that it is cloudy 80% of the time (sometimes it is cloudy without rain). You know, of course, that 100% of the time, if it is raining, then it is also cloudy. What do you think the chances are of rain, given that it is just cloudy? Well, Bayes rule allows you to compute this. Bayes rule says that
p(R|C) = p(R)p(C|R)/p(C) = 0.5 x 1.0 / 0.8 = 0.625 = 5/8.
So, 5/8 of the time, in London during winter, if it is cloudy, then it is rainy.
Bayes rule can be extended to multiple variables with multiple states, which requires automated tools to be solved efficiently.

Bayes Rule:
For any two events, A and B,
p(B|A) = p(A|B) x p(B) / p(A)

where you read 'p(A)' as "the probability of A", and 'p(A|B)' as "the probability of A given that B has occurred".

UNIVERSITÀ DEGLI STUDI FIRENZE

How does a BN work



False	0.5
True	0.5

A	False	True
False	0.5	0.3
True	0.5	0.7

Bayes Theorem

$$P(B \mid A) = \frac{P(A \mid B) \times P(B)}{P(A)}$$

Joint probability of all variables can be computed as follows:

$$P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \text{Parents}(X_i))$$

UNIVERSITÀ DEGLI STUDI FIRENZE

Why «acyclic»?

- Suppose we have three variables A, B, and C, and assume the graph has a **cycle** like this:

  $A \rightarrow B \rightarrow C \rightarrow A$ (each variable depends on the next, and C depends back on A).

- Now, in a Bayesian network (normally **acyclic**), the joint probability would be factored like:

  $P(A,B,C)=P(A)\times P(B|A)\times P(C|B)$ (no problem here — **no cycles**).

- But **with the cycle**, how would you factor P(A,B,C)? You'd want to write:

  $P(A)\times P(B|A)\times P(C|B)\times P(A|C)$ because A now depends on C.

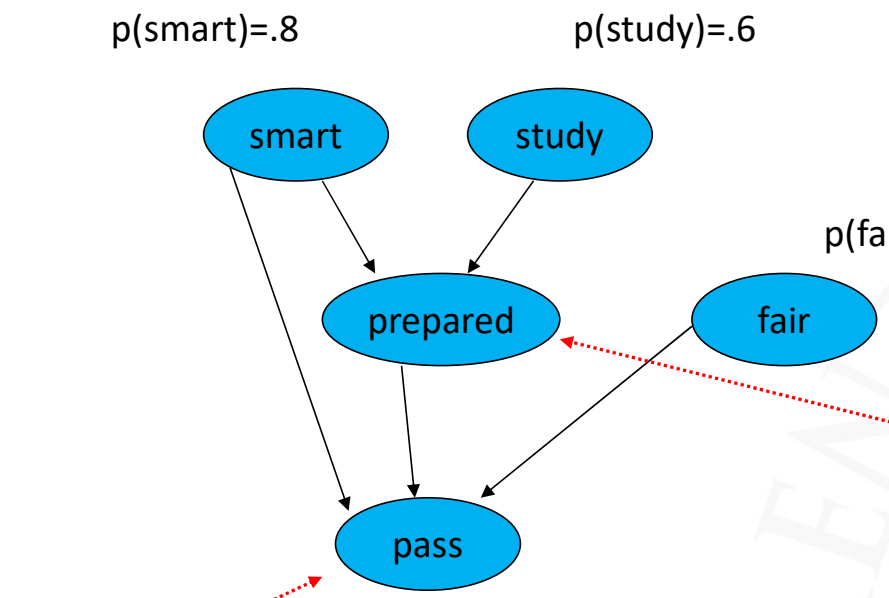
- **Problem**: You're trying to define P(A) based on C, which in turn is based on B, which is based on A... so you are **defining A in terms of itself**. You're **stuck in a loop** — no clear starting point. As a result: The factorization fails. The probability space might not even be valid (probabilities might not sum to 1). Inference algorithms would get into infinite recursion or never converge.

UNIVERSITÀ DEGLI STUDI FIRENZE

BN modeling and evaluation tool

- Several tools available
- We use *Norsys Netica* (https://www.norsys.com/index.html) for teaching purposes
- Free version has limitations but it is enough for most simple applications
- Download from: https://www.norsys.com/download.html
- Tutorial available at: https://www.norsys.com/tutorials/netica/nt_toc_A.htm

BN example: probability to pass the exam



p(prep|…)	smart	¬smart
study	.9	.7
¬study	.5	.1

p(pass|…)	smart		¬smart	
	prep	¬prep	prep	¬prep
fair	.9	.7	.7	.2
¬fair	.1	.1	.1	.1

UNIVERSITÀ DEGLI STUDI FIRENZE

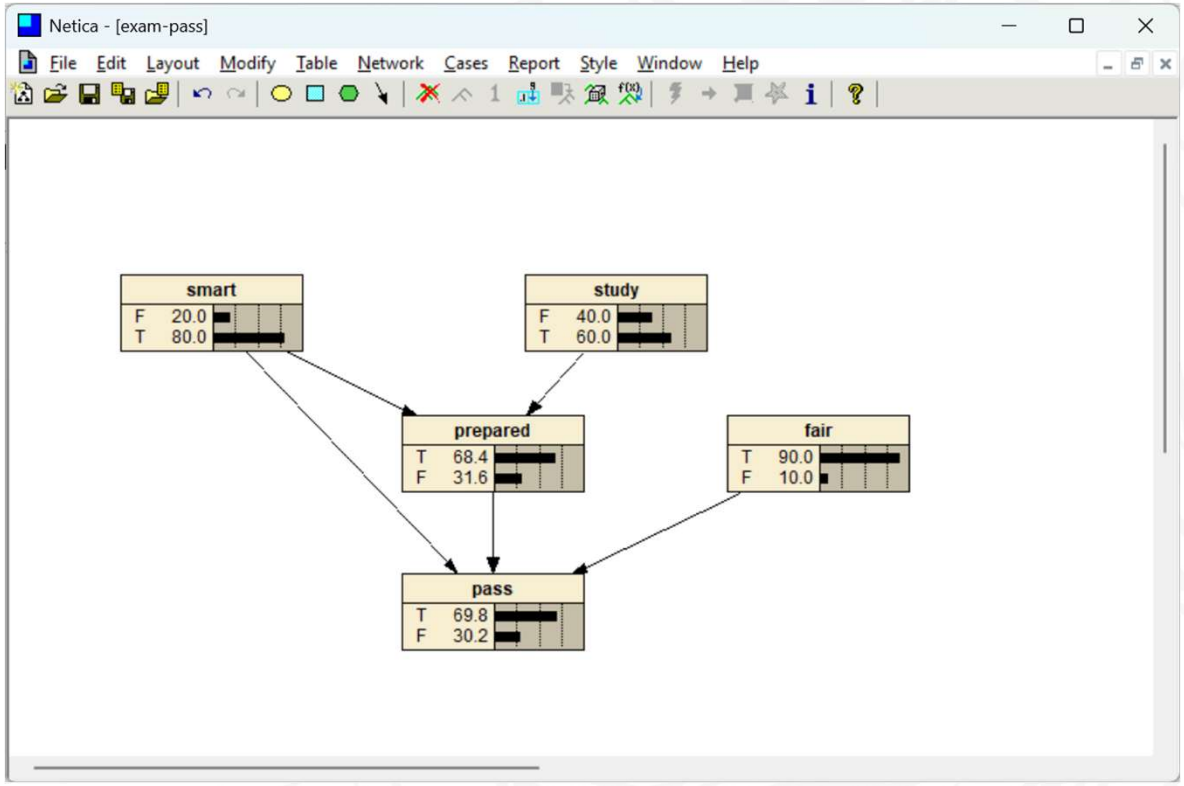
Direct inference

Query: What is the probability that a student passes the exam?

UNIVERSITÀ DEGLI STUDI FIRENZE

Direct inference result in Netica

"Network" -> "Compile"



UNIVERSITÀ DEGLI STUDI FIRENZE

Most Probable Explanation

Goal: Find the single most likely assignment to all unobserved variables, given evidence.

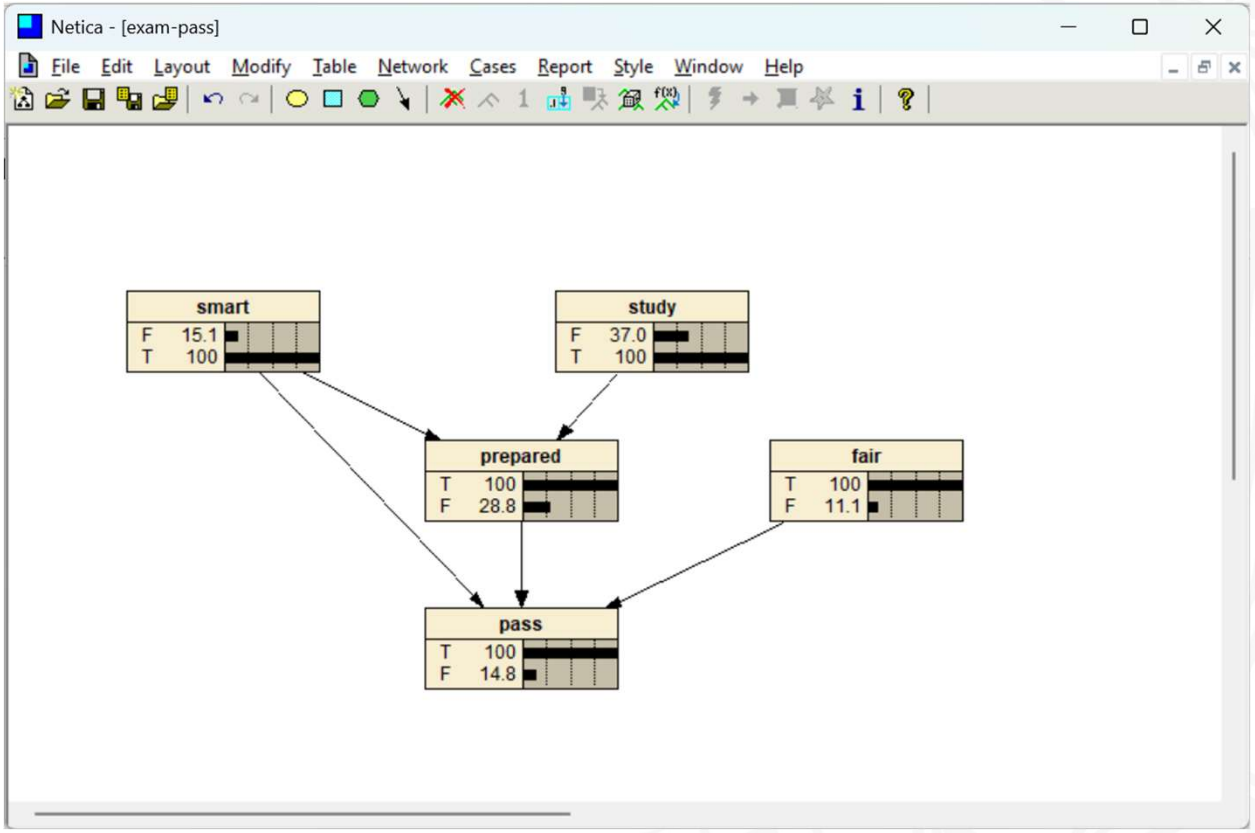
Output: A complete assignment (one scenario), not a distribution.

Example exam-pass BN query:

What is the most likely configuration of events for successfully passing the exam?

UNIVERSITÀ DEGLI STUDI FIRENZE

Most Probable Explanation



UNIVERSITÀ DEGLI STUDI FIRENZE

A-posteriori analysis

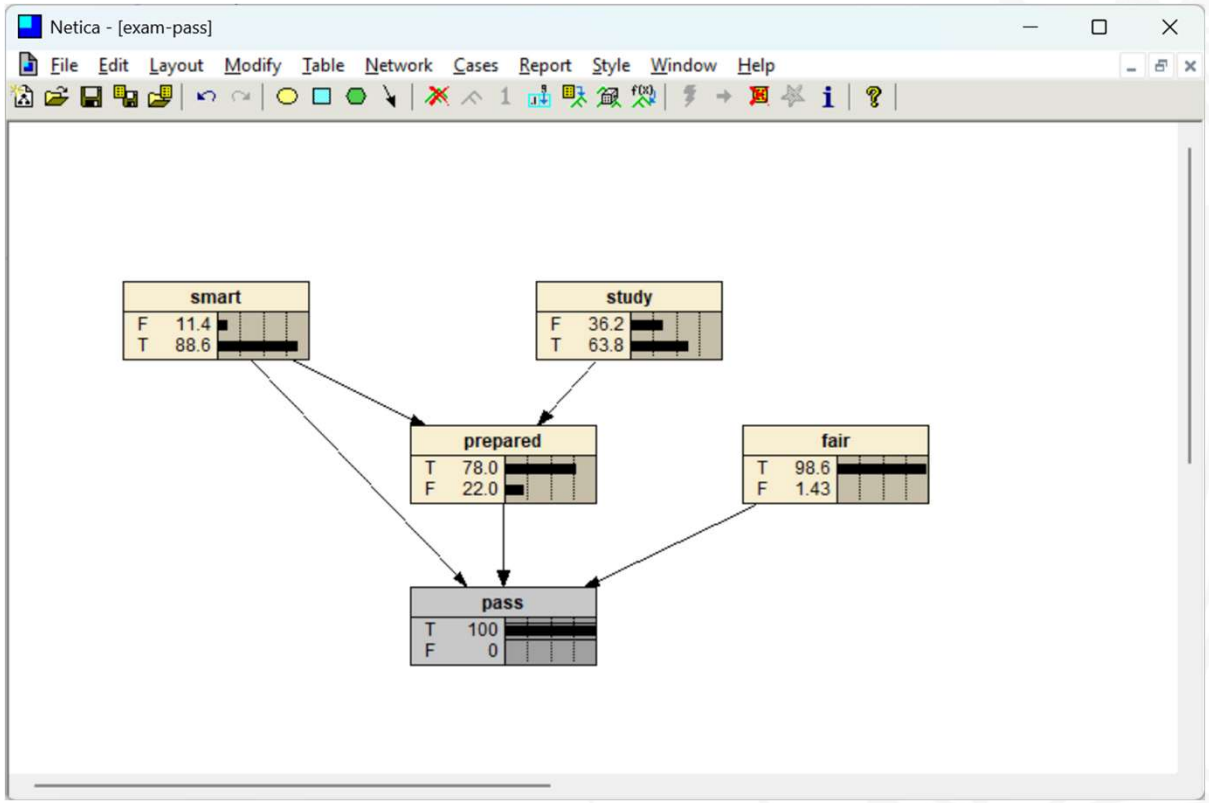
Goal: Compute the probability distribution of one or more unobserved variables, given evidence.

Output: A probability distribution (not a single value).

This is useful when you want **uncertainty quantification**: not just the most likely cause, but how likely each possible cause is.

UNIVERSITÀ DEGLI STUDI FIRENZE

A-posteriori analysis result in Netica for the exam-pass BN



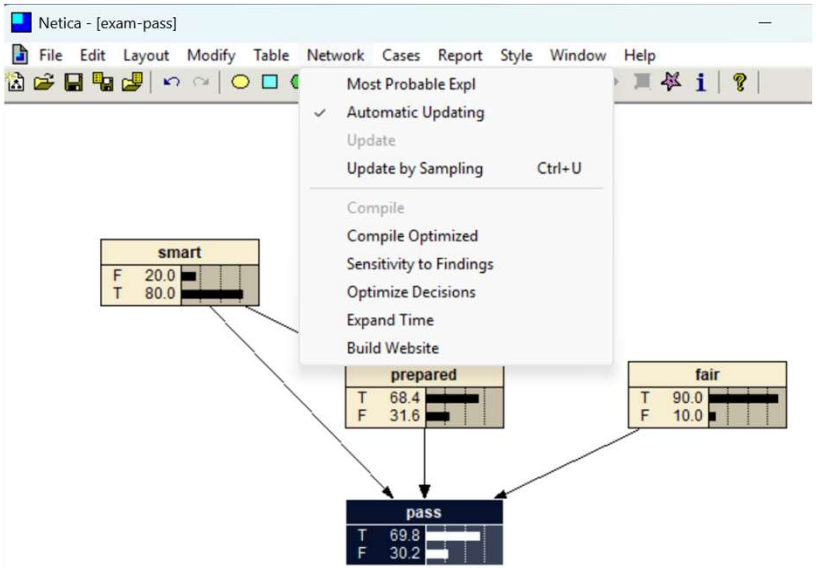
UNIVERSITÀ DEGLI STUDI FIRENZE

Sensitivity to findings

- **Sensitivity to findings** in a BN refers to **how much a change in the observed evidence (findings)** affects the **beliefs (posterior probabilities)** of other BN variables.
- It answers the question: "If I change or remove this piece of evidence, how much do my conclusions change?"
- This is useful for:
  - Understanding how influential a certain observation is.
  - Detecting critical evidence or redundant information.
  - Performing robustness analysis of your model.

UNIVERSITÀ DEGLI STUDI FIRENZE

Sensitivity to findings: exam-pass BN example




```
Sensitivity of 'pass' to a finding at another node:

Node                    Mutual      Percent   Variance of
----                    Info                  Beliefs
pass                    0.88377     100       0.2108118
fair                    0.12799     14.5      0.0397285
smart                   0.07202     8.15      0.0226442
prepared                0.06844     7.74      0.0207161
study                   0.01013     1.15      0.0029891
```

UNIVERSITÀ DEGLI STUDI FIRENZE

«Noisy» logic gates in BN

Noisy-gates are **probabilistic generalizations of logical gates (AND, OR, etc.)** used in Bayesian networks to model **uncertain causal relationships** between binary variables.

In real-world systems:

- Causes do **not always** deterministically lead to effects.
- Each cause might succeed or fail **independently** in triggering an effect.

Noisy-gates allow us to model this kind of **imperfect, probabilistic logic** efficiently — especially when multiple parent nodes influence a child.

UNIVERSITÀ DEGLI STUDI FIRENZE

Example: Noisy-AND

- The **noisy-AND** is a common and powerful **probabilistic model** used in Bayesian networks when multiple causes can independently contribute to an effect — but **not deterministically**. It generalizes the classical **logical AND** gate to allow for **uncertainty** or **failure** in causal relationships.
- In a **pure logical AND**, the effect happens *only if all causes are present* — and it happens with certainty. But real-world systems often behave **probabilistically**:

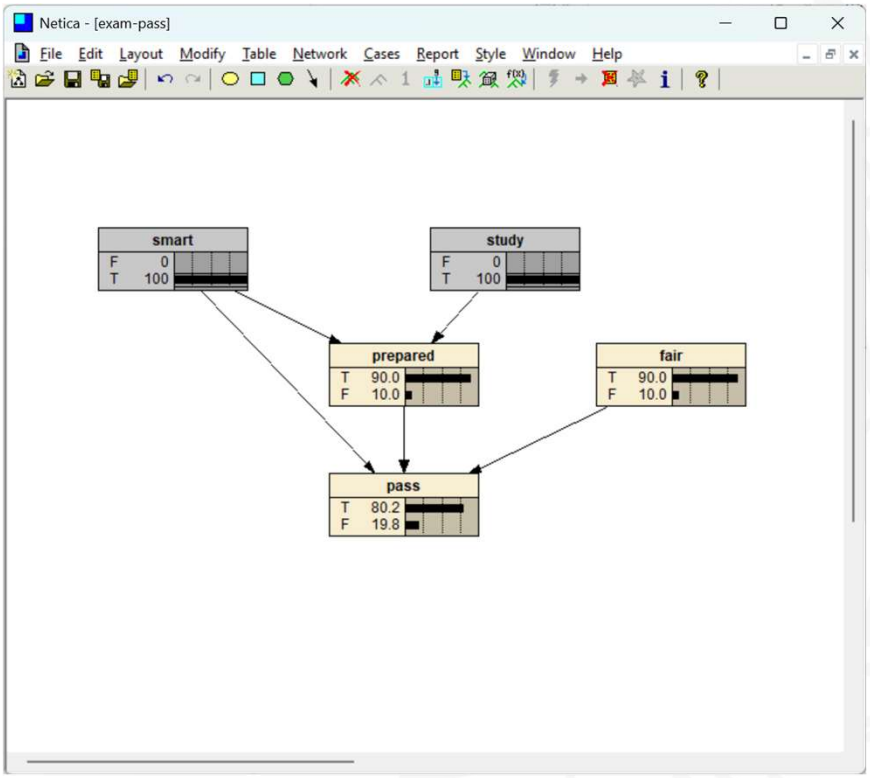
  > "Even if all the required causes are present, the effect might still not happen (due to noise, failure, randomness)."

This is where noisy-AND comes in.

Similarly with **noisy-OR**.

UNIVERSITÀ DEGLI STUDI FIRENZE

Noisy-AND example in the exam-pass BN



UNIVERSITÀ DEGLI STUDI FIRENZE

How efficient is inference in BNs?

- Question relevant for scalability to large systems and/or complex problems

- Short answer: it depends…

- Generally, **exact inference in arbitrary BNs is NP-hard**.
- But **it can be efficient (Polynomial Time) if the graph is tree-structured**.

UNIVERSITÀ DEGLI STUDI FIRENZE

Classroom assignment and discussion

- Sketch a simple BN representing a situation of your choice (individual). Define the relevant variables and connections between them. Try to assign a-priori probabilities to evidence nodes and quantify the CPT.

- You can use pencil and paper, or your laptop, if you have one with you.

- After you finish your first draft, start explaining and discussing the model and its parameters with your classmate sitting next to you.

UNIVERSITÀ DEGLI STUDI FIRENZE

Home assignment

- If you did not do it during the lecture, download and install Netica.
- Implement your first model in Netica and start experimenting with inference, most probable explanations, a-posteriori analyses, and sensitivity to findings.
- Prepare a few slides Powerpoint presentation about your model and results to be possibly shown and discussed during next lecture.
- Such assignment is for learning purposes only, and therefore not graded.
- Working in small groups is allowed.

UNIVERSITÀ DEGLI STUDI FIRENZE

Next lecture

- Quick recap and questions about today's lecture

- Presentation of basic BN models developed by students

- We will start investigating dependability modeling applications, showing transformation from Fault Trees to Bayesian Networks

UNIVERSITÀ DEGLI STUDI FIRENZE

Thank you for your attention

Questions?

Prof. Francesco Flammini

francesco.flammini@unifi.it